TASK 2 – PRODUCT & CATEGORY MANAGEMENT

SQL Database Documentation

Database: BrightMartDB

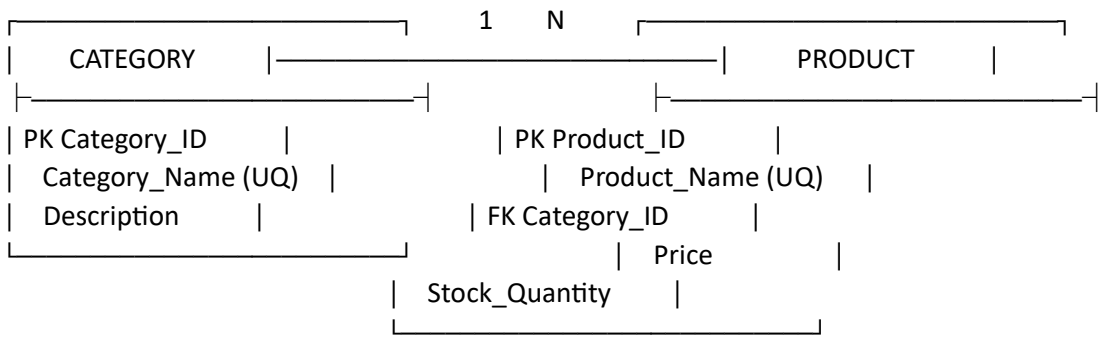
1. Category & Product Table Design

The database contains two related tables: Category stores product categories, while Product stores individual products. Category_ID is used as the primary key in Category and as a foreign key in Product.

Table	Column	Data Type	Constraints
Category	Category_ID	INT	PRIMARY KEY, AUTO_INCREMENT
Category	Category_Name	VARCHAR(50)	NOT NULL, UNIQUE
Category	Description	VARCHAR(150)	Optional
Product	Product_ID	INT	PRIMARY KEY, AUTO_INCREMENT
Product	Product_Name	VARCHAR(100)	NOT NULL, UNIQUE
Product	Category_ID	INT	NOT NULL, FOREIGN KEY
Product	Price	DECIMAL(10,2)	NOT NULL, CHECK (Price > 0)
Product	Stock_Quantity	INT	NOT NULL, CHECK (Stock_Quantity >= 0)

2. ER Diagram

Relationship: One Category can contain many Products (1 : N).



3. SQL Table Creation & Constraints

```
CREATE DATABASE BrightMartDB;
USE BrightMartDB;
```

```
CREATE TABLE Category (
   Category_ID INT PRIMARY KEY AUTO_INCREMENT,
   Category_Name VARCHAR(50) NOT NULL UNIQUE,
   Description VARCHAR(150)
);

CREATE TABLE Product (
   Product_ID INT PRIMARY KEY AUTO_INCREMENT,
   Product_Name VARCHAR(100) NOT NULL UNIQUE,
   Category_ID INT NOT NULL,
   Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
   Stock_Quantity INT NOT NULL CHECK (Stock_Quantity >= 0),
   FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
);
```

4. Sample Category Data

Category_ID	Category_Name	Description
1	Electronics	Electronic devices and accessories
2	Home Appliances	Appliances used at home
3	Fashion	Clothing and fashion products

```
INSERT INTO Category (Category_Name, Description) VALUES
('Electronics', 'Electronic devices and accessories'),
('Home Appliances', 'Appliances used at home'),
('Fashion', 'Clothing and fashion products');
```

5. Sample Product Data

Product_ID	Product_Name	Category_ID	Price	Stock_Quantity
1	Smartphone	1	24999.0	25
2	Laptop	1	55999.0	15
3	Bluetooth Speaker	1	2499.0	40
4	Wireless Mouse	1	799.0	60
5	Smart Watch	1	3499.0	30
6	Refrigerator	2	32999.0	10
7	Mixer Grinder	2	4499.0	20
8	Air Conditioner	2	42999.0	8
9	T-Shirt	3	699.0	50
10	Jeans	3	1499.0	35

```
INSERT INTO Product (Product_Name, Category_ID, Price, Stock_Quantity) VALUES
('Smartphone', 1, 24999.00, 25),
('Laptop', 1, 55999.00, 15),
('Bluetooth Speaker', 1, 2499.00, 40),
('Wireless Mouse', 1, 799.00, 60),
('Smart Watch', 1, 3499.00, 30),
('Refrigerator', 2, 32999.00, 10),
('Mixer Grinder', 2, 4499.00, 20),
('Air Conditioner', 2, 42999.00, 8),
('T-Shirt', 3, 699.00, 50),
('Jeans', 3, 1499.00, 35);
```

6. CRUD Operations

```
-- INSERT
INSERT INTO Product (Product_Name, Category_ID, Price, Stock_Quantity)
VALUES ('Keyboard', 1, 1299.00, 25);

-- SELECT
SELECT * FROM Product;

-- UPDATE
UPDATE Product
SET Price = 1199.00, Stock_Quantity = 30
WHERE Product_Name = 'Keyboard';

-- DELETE
DELETE FROM Product
WHERE Product_Name = 'Keyboard';
```

7. Category-wise Analysis Queries

```
-- 1. Display products category-wise
SELECT c.Category_Name, p.Product_Name, p.Price, p.Stock_Quantity
FROM Category c
JOIN Product p ON c.Category_ID = p.Category_ID
ORDER BY c.Category_Name, p.Product_Name;

-- 2. Count products in each category
SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
FROM Category c
LEFT JOIN Product p ON c.Category_ID = p.Category_ID
GROUP BY c.Category_ID, c.Category_Name;

-- 3. Find highest-priced product
SELECT Product_Name, Price
FROM Product
WHERE Price = (SELECT MAX(Price) FROM Product);

-- 4. Find categories having more than 5 products
```

```
SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
FROM Category c
JOIN Product p ON c.Category_ID = p.Category_ID
GROUP BY c.Category_ID, c.Category_Name
HAVING COUNT(p.Product_ID) > 5;

-- 5. Calculate average product price
SELECT AVG(Price) AS Average_Product_Price
FROM Product;
```

8. Category-wise Analysis Report

Category	Product Count	Highest Price	Average Price
Electronics	5	₹24,999.00	₹17,559.00
Home Appliances	3	₹42,999.00	₹26,809.00
Fashion	2	₹1,499.00	₹1,099.00

From the sample data, Electronics contains 5 products, Home Appliances contains 3 products, and Fashion contains 2 products. The highest-priced product is the Air Conditioner at ₹42,999.00. No category has more than 5 products in the initial sample data.

9. Expected Query Outputs

Query	Expected Result
Highest-priced product	Air Conditioner — ₹42,999.00
Overall average price	₹17,139.10
Category with > 5 products	No rows for the initial 10-product sample

10. Submission Checklist

✓ Category & Product table design

✓ ER diagram

✓ Sample product data (10 products)

✓ Primary Key, Foreign Key, NOT NULL, UNIQUE and CHECK constraints

✓ CRUD operations: INSERT, SELECT, UPDATE, DELETE

✓ Category-wise analysis queries

✓ Expected query output summary